Tanvir Chowdhury

Embedded Systems Engineer
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github.com/tanvirc | scholar.google.com/citations?user=TnVr0Q1

Summary

Embedded engineer designing low-power sensor hardware and firmware for agriculture.

Experience

Krishi Sense Labs · Kolkata
Co-founder and Hardware Lead Jun 2021 – Present

- Designed a soil-moisture node that runs 3 years on two AA cells.
- Raised a seed grant from the state startup mission.
- Hired and mentor a team of five engineers.
- Ship firmware over the air to 4,000 deployed devices.

Texas Instruments · Bengaluru
Embedded Software Engineer Jul 2018 – May 2021

- Wrote drivers for the CC13xx sub-GHz radio family.
- Reduced idle current of the reference design by 38 percent.
- Maintained the BLE stack examples used by customers.

ISRO Satellite Centre · Bengaluru
Project Trainee Dec 2017 – May 2018

- Tested telemetry decoders for a student satellite payload.

Education

B.E. in Electronics and Telecommunication Engineering
Jadavpur University · Jul 2014 – May 2018 · CGPA 9.02/10

Class XII
South Point High School · Mar 2014 · 94.2%

Selected Projects

PocketScope

- Two-channel USB oscilloscope on a single four-layer board.
- Streams 1 MS/s to a web front end over WebUSB.

Tech: C, STM32, KiCad
GitHub: github.com/tanvirc/pocketscope

Irrigation Mesh Controller

- Firmware for valve controllers that coordinate over a LoRa mesh.
- Survives brown-outs without corrupting the schedule.

Tech: Zephyr RTOS, LoRa, Rust

Patents and Publications

- "Low-power soil moisture sensing apparatus with capacitive guard ring." *Indian Patent Application 202131045678*, 2022.
- T. Chowdhury, R. Sen. "Energy-aware duty cycling for LoRa sensor meshes." *IEEE Sensors Journal*, 2023.

Scholarships and Honours

- KVPY Fellowship — Department of Science and Technology, 2013
- University Gold Medal — Jadavpur University, 2018

Certifications

- Embedded Systems Essentials with Arm · Arm Education · Issued Apr 2020 · ID ARM-ESE-55120

Leadership

IEEE Student Branch, Jadavpur University
Chair Aug 2016 – Jul 2017

- Ran a 300-participant hardware hackathon.

Volunteering

Robotics for Rural Schools
Mentor Jan 2019 – Present

- Teach weekend electronics workshops.

Technical Skills

Firmware: C, C++, Rust, Zephyr RTOS, FreeRTOS
Hardware: KiCad, STM32, LoRa, BLE

Languages

Bengali — Native
English — Fluent
Hindi — Fluent

Interests

- Amateur radio (VU2TNV)
- Long-distance cycling